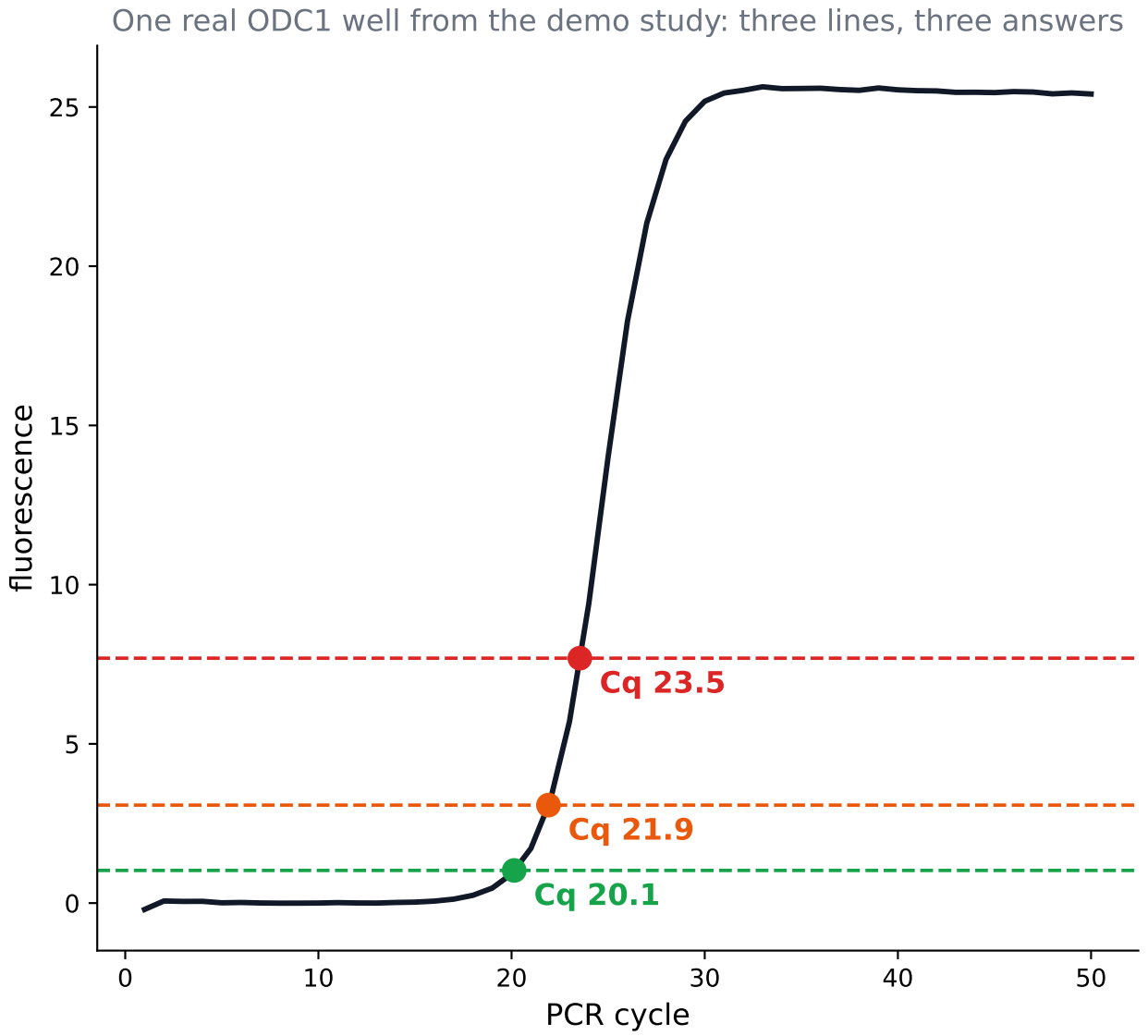
qPilot

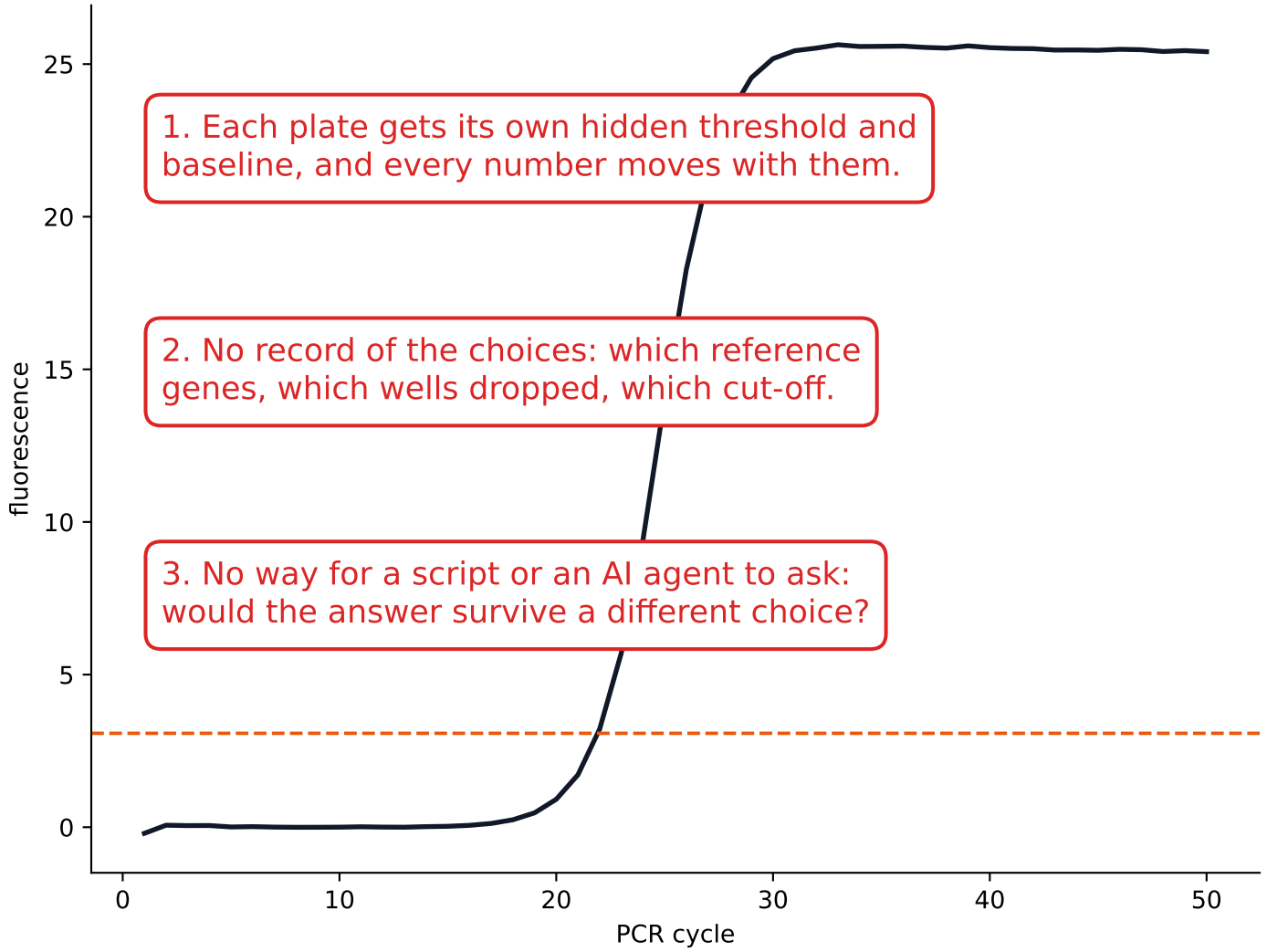
Every qPCR result rests on where one line is drawn.

Nobody writes down where.

Open source · runs on your laptop · research use only



What wasn't working



5 % of papers validate their reference genes

1 % report amplification efficiency

#4 lab-instrument software, on this hackathon's own survey

Bustin & Wittwer, Clinical Chemistry, 28 Jan 2026, doi:10.1093/clinchem/hvaf176 (355 papers)

What qPilot does

One Reads the raw curves of all 64 plates (24,576 reactions) and analyses every plate with the same published method and settings, written down.

Two Finds the 2 choices that could change a conclusion. A person decides each one, and the reason goes in the record.

Three Tests every conclusion under 186 reasonable analyses: ODC1 holds in 100%, NM23A in 65%. Any AI can ask, with numbers it cannot invent.

Card 1, the reference genes: 31 of 58 conclusions change; 17 even between sets of three or more genes. The HMBS:UBC(2) ratio itself shifts 1.8× between the groups (p = 0.058).

Agent clip: qpilot serve (MCP), shown live · Backup video: https://github.com/ShivaTeja05/Qpilot/blob/main/demo/demo.mp4

Proof

Validation

Reproduces the RDML consortium's published benchmark on 930 standard wells: 99.6 % identical, 100 % within 5 %; per-gene bias matches to a median 0.02; a +0.97-cycle Cq shift is disclosed. Data: Vermeulen 2009 / Ruijter 2013, CC BY 4.0; groups by MYCN expression.

Engineering

398 tests · CI green, pip-audit clean · one-command install · byte-identical re-run (tested) · THREAT_MODEL.md

Security

Offline by test: the whole analysis runs with the network switched off. The app binds 127.0.0.1, telemetry off. An AI sees the facts and the card labels, never the raw data; the app shows exactly what it was sent.

Cost

Today: €2.4–6k/yr per academic lab for the analysis-and-stats step (Clarida Lab €4,800–9,600 as listed; Prism $260–520/seat). qPilot: Apache-2.0, laptop. (Does not replace the instrument software, Part 11 editions, or Prism for non-qPCR work.)

For research use only. Not for clinical or diagnostic use.

github.com/ShivaTeja05/Qpilot

Automatic: a choice that cannot change any conclusion is decided by the system and logged.

Agent may apply: such non-flipping choices, when the scientist allows decide-and-tell.

Human only: any choice that can flip a conclusion, at every level, for every actor.

Built: the decision space, robustness grid, cards, record, app, MCP, model harness. Borrowed: LinRegPCR (rdmlpython, MIT), geNorm, RDML, the demo data.